

1.

$$(1) \quad I(t) = \frac{dq(t)}{dt} = 10\pi f_1 \cos(2\pi f_1 t) - 4\pi f_2 \sin(2\pi f_2 t) = 10\pi \cos(2000\pi t) - 12\pi \sin(6000\pi t) \text{ (mA)}$$

(2) 当 $t = 50\text{ms}$ 时,

$$I(t) = 10\pi \cos(100\pi) - 12\pi \sin(300\pi) = 10\pi = 31.416 \text{ (mA)}$$

2.

$$(1) \quad V_{rms} = \frac{W}{I_{rms} t} = \frac{216 \times 10^3}{270 \times 10^{-3} \times 3600} = 222.22 \text{ (V)}$$

$$(2) \quad R = \frac{V_{rms}}{I_{rms}} = 1058.20 \text{ (}\Omega\text{)}$$

$$(3) \quad W = 216 \text{ (kJ)} = \frac{216 \times 10^3}{3600} \text{ (Wh)} = 60 \text{ (Wh)}$$

3.

(1)

$$(a) \quad P = vi = (1000 \times 50 \cos(100\pi \times 3 \times 10^{-3}) \times 10^{-3}) \times (50 \cos(100\pi \times 3 \times 10^{-3}) \times 10^{-3}) = 0.864 \text{ (W)}$$

$$(b) \quad P = 0.003 \times (-5000\pi \sin(0.3\pi)) \times 10^{-3} \times 0.05 \cos(0.3\pi) = -1.120 \text{ (W)}$$

$$(c) \quad P = 10 \times 50 \times \cos(0.3\pi) \times 10^{-3} + 10000 \left(\int_0^{0.003} 50 \cos(100\pi t) dt \right) \times 10^{-3} \times 50 \times \cos(0.3\pi) \times 10^{-3} = 0.331 \text{ (W)}$$

(2) (把所有的单位都化成国际单位)

$$(a) \quad \bar{P} = \lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T vi dt = \lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T (2.5 \cos^2(100\pi t)) dt$$

$$= \lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T (1.25 + 1.25 \cos(200\pi t)) dt = 1.25 \text{ (W)}$$

$$(b) \quad \bar{P} = \lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T vi dt = \lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T 0.003 i di = 0.003 \lim_{T \rightarrow +\infty} \frac{1}{T} [0.05 \cos^2(100\pi t)]_0^T = 0 \text{ (W)}$$

$$(c) \quad v = 10 + 10000 \int_0^t i dt \text{ (mV)} = 0.01 + 10 \int_0^t 50 \cos(100\pi t) dt \text{ (V)} = 0.01 + \frac{5}{\pi} \sin(100\pi t) \text{ (V)}$$

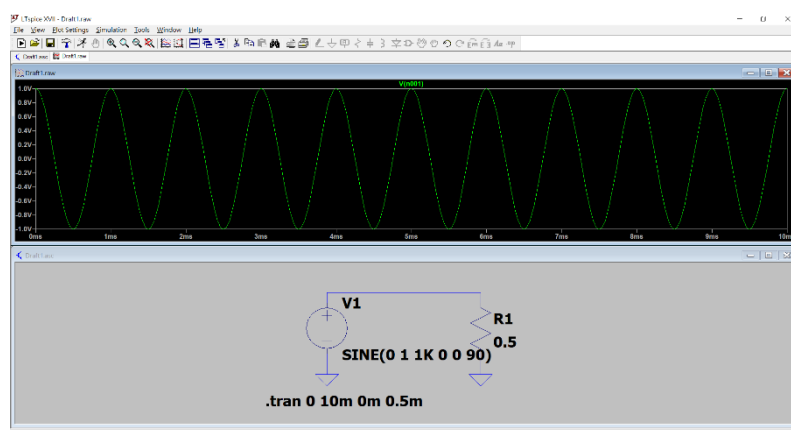
$$\bar{P} = \lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T vi dt = \lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T \left(5 \times 10^{-4} \cos(100\pi t) + \frac{1}{8\pi} \sin(200\pi t) \right) dt = 0 \text{ (W)}$$

(3) 电阻是无源元件，电感和电容是有源元件。

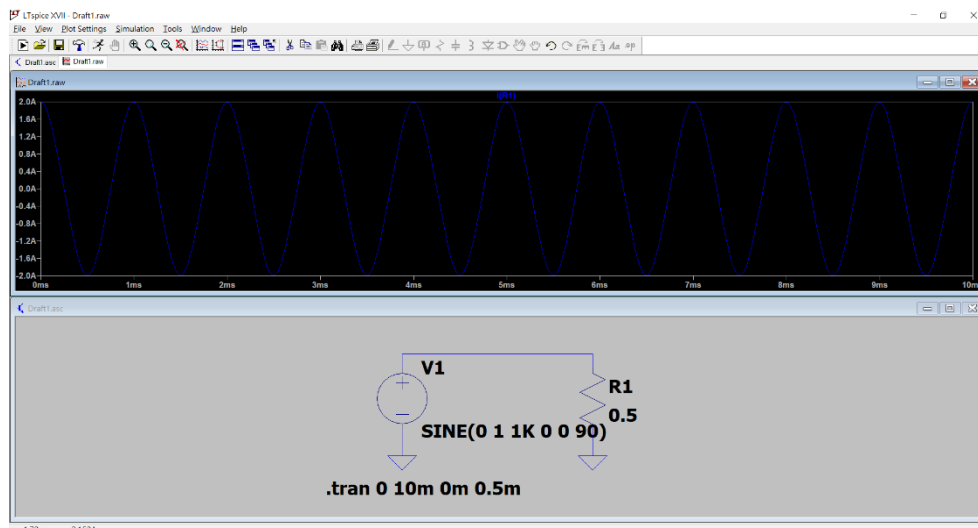
4.

(1) 线性时不变电阻：

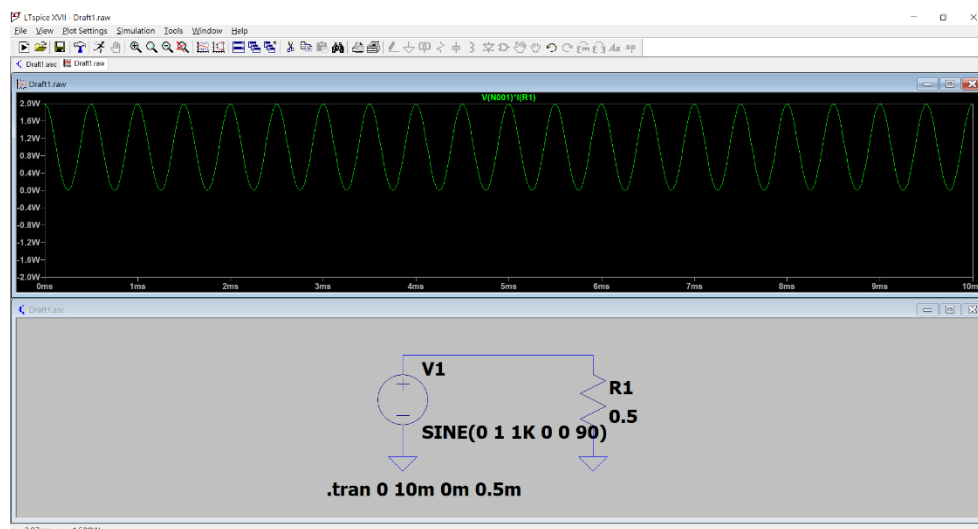
(I.) 电压波形：



(II.) 电流波形:



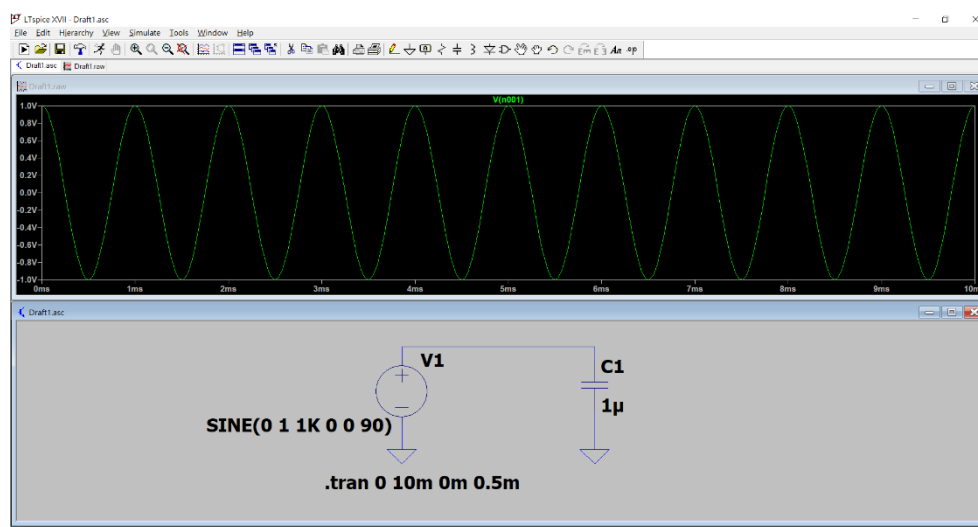
(III.) 功率波形:



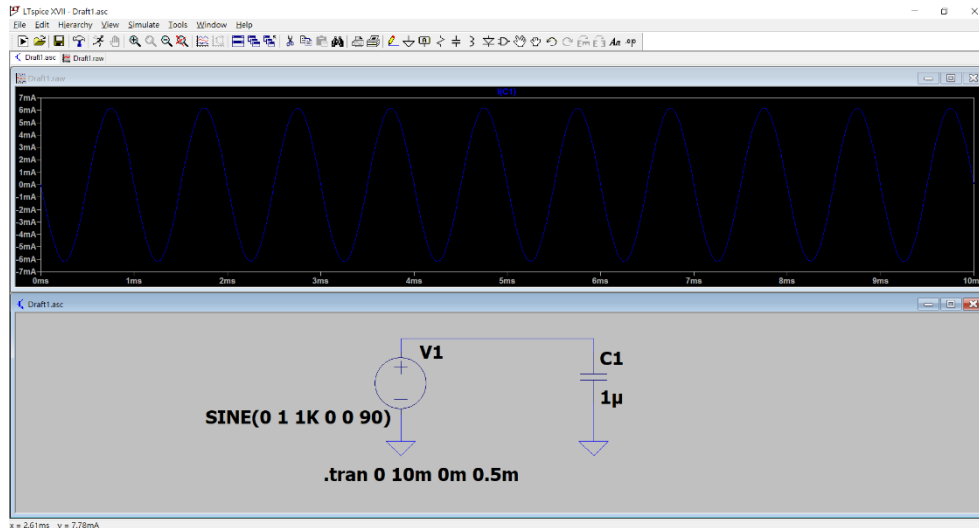
可以观察到, 功率 $P \geq 0$, 所以电阻只能吸收电能, 故是耗能元件。

(2) 线性时不变电容:

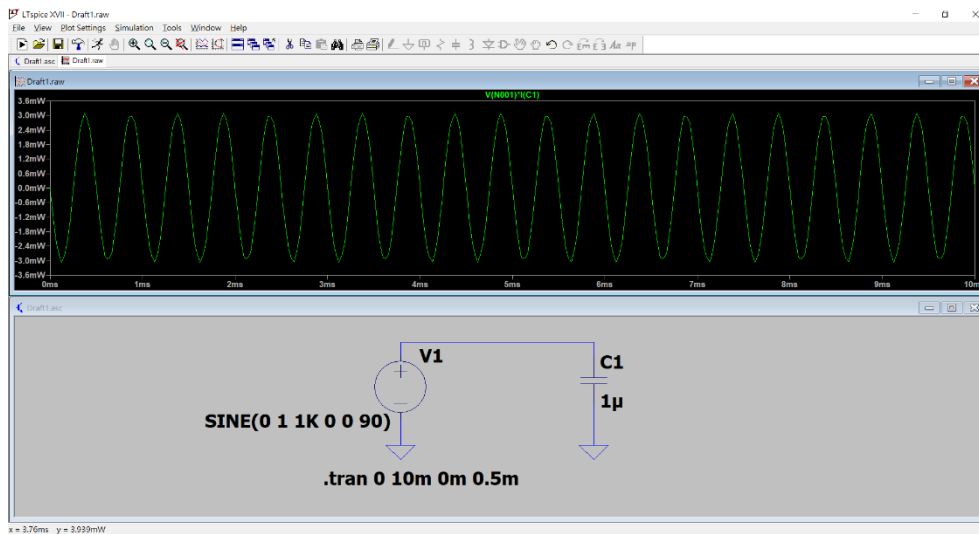
(I.) 电压波形:



(II.) 电流波形:



(III.) 功率波形:

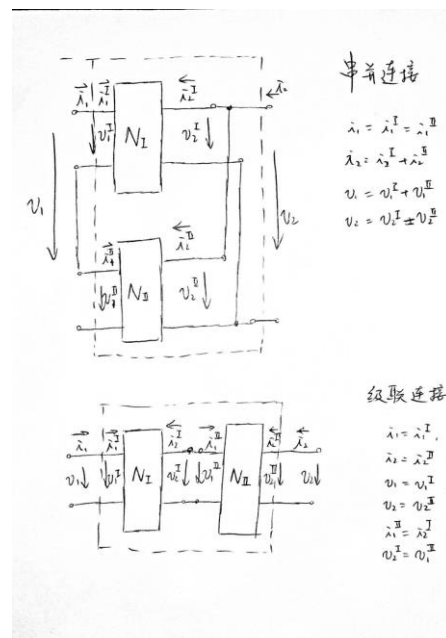
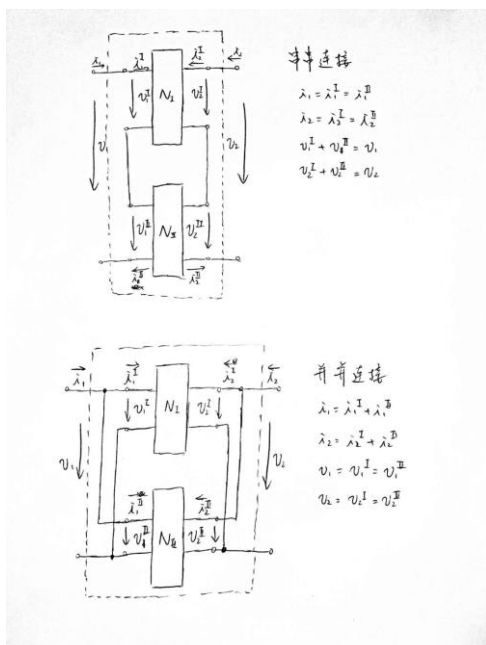


由功率波形可见，波形关于 $P = 0$ 轴对称，而且存在 $P < 0$ 的区间，所以电容是储能且可以释放能量。

5. 驱动电压 $v = \frac{P}{I} = \frac{P}{\frac{dq}{dt}} = \frac{10}{1.6 \times 10^{-19} \times 10^{15}} 6 = 6.25 \times 10^4 (V) = 62.5 (kV)$

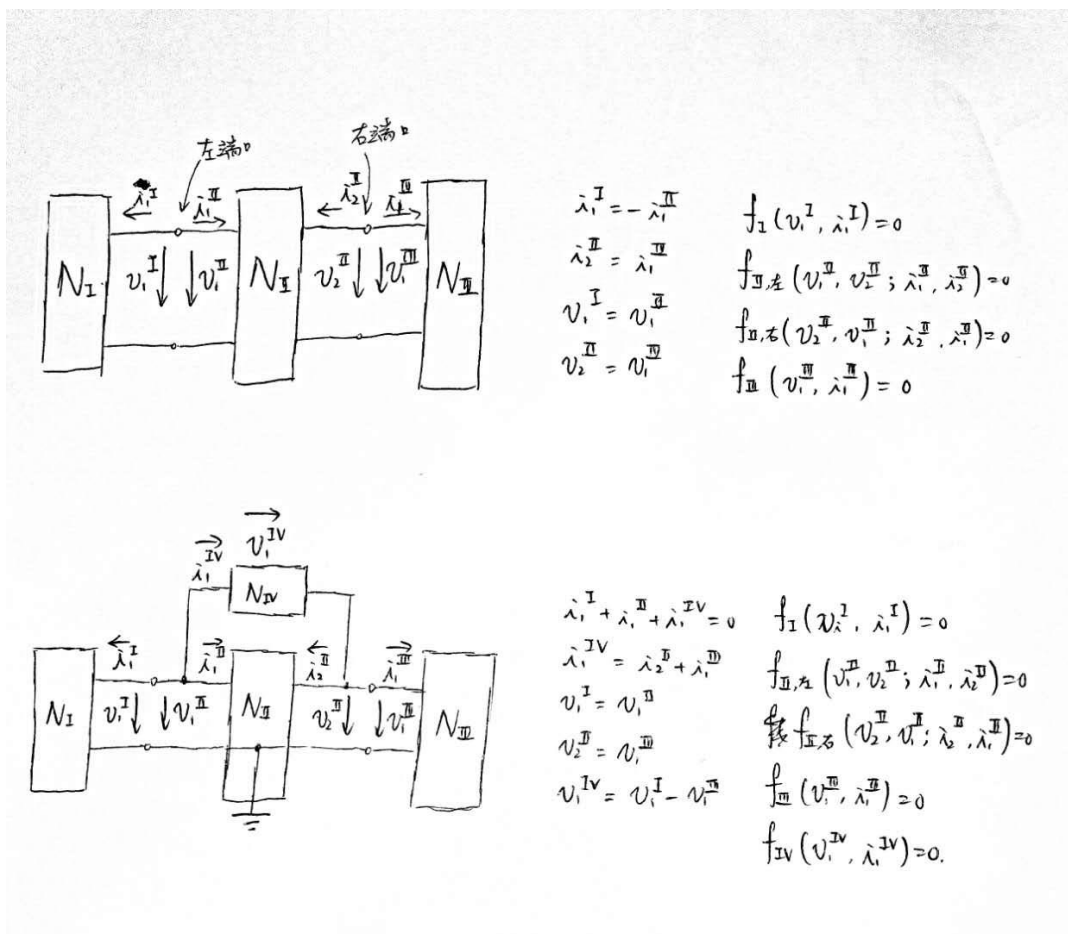
6.

(1) 和 (2) 的答案:



(3) 当两个压控网络并联时, $i = i^I + i^{II} = f_I(v_I) + f_{II}(v_{II}) = f_I(v) + f_{II}(v) = f(v)$
仍然是压控网络。

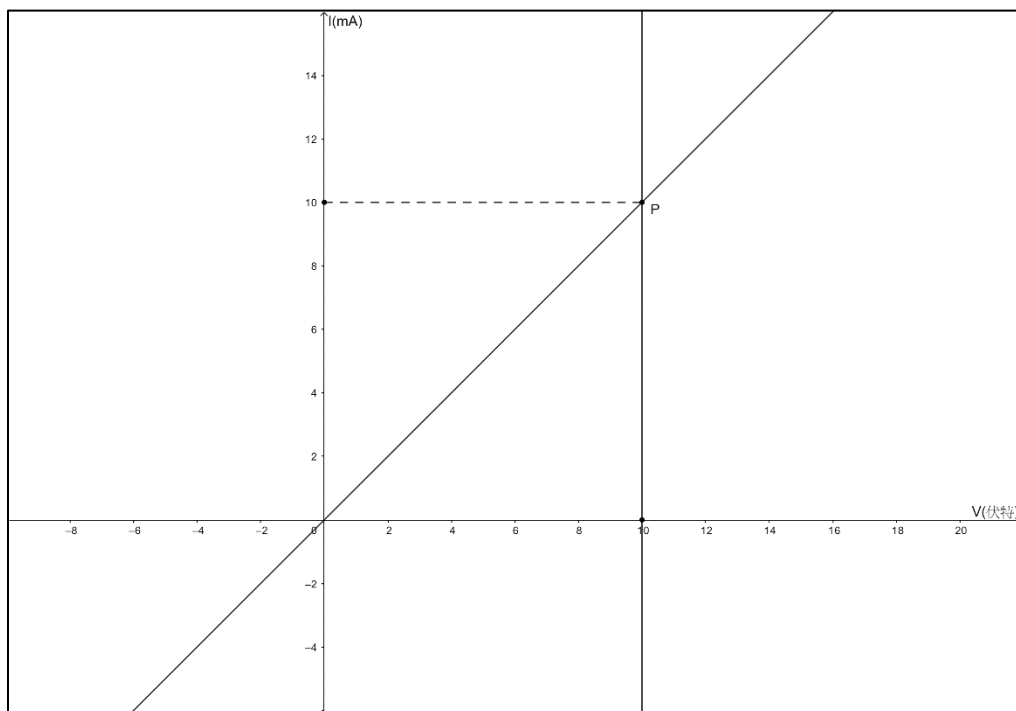
7.



8.

(1) 直流电压源:

$$v = V_{so} = 10(V), \quad i = \frac{v}{R_L} = 0.01(A) = 10(mA)$$



(2) 交流电压源:

$$v = V_{so} = 10 \cos(\omega t) (V), \quad i = \frac{v}{R_L} = 10 \cos(\omega t) (mA)$$

